

Notes: Gupta, Kopytov & Starmans (RFS, 2026)
The Pace of Change: Socially Responsible Investing in Private Markets

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1 Big picture

- **Question.** Can socially responsible investors (SRIs) speed up firm reform in private markets, or can their presence unintentionally slow the pace at which dirty firms become green?
- **Core mechanism.** SRIs with broad prosocial preferences value acquiring dirty firms because they can reform them. That willingness to pay creates a resale option for current owners of dirty firms, weakening the owners' incentive to reform proactively.
- **Main paradox.** Investors who care most about reducing externalities may increase the value of staying dirty. This can generate *strategic delay*: the owner delays reform because she expects an SRI to pay for a dirty target later.
- **Private-market focus.** The model is built around search frictions and acquisition bargaining, making it especially relevant for impact funds, private equity, venture capital, private credit, and other negotiated-control settings.
- **Role of financial investors.** Financial investors (FIs) do not care about externalities. Yet their presence can worsen delay because they can create fallback trading and resale options for dirty firms.
- **Mandate result.** SRIs can fix the delay problem by committing ex ante to pay a premium for green firms. This changes incentives before acquisition occurs.
- **Measurement implication.** Impact may occur *before* the SRI invests. If a green-premium mandate induces the seller to reform before sale, post-investment ESG changes understate the investor's impact.
- **Policy implication.** Sustainable-finance contracts matter. Active acquisition of dirty firms is not always faster than passive commitment to reward already-green firms.

2 Data, objects, and economic environment

This is a theoretical paper. The relevant objects are model primitives, states, timing assumptions, payoffs, and equilibrium thresholds rather than empirical datasets.

2.1 Agents

- An **entrepreneur** initially owns a dirty firm.
- There are N potential investors.
- N_f are **financial investors**; $N_s = N - N_f$ are **socially responsible investors**.
- FIs value financial profits only. SRIs value profits and dislike the firm's social cost.
- All agents are risk neutral and share discount factor $\beta \in (0, 1)$.

Interpretation: The investor distinction is central. FIs create financial outside options; SRIs create social surplus by reforming the firm. The interaction of these two roles generates the timing distortion.

2.2 Firm types and reform

- Each period, the firm generates profit $\pi > 0$.
- A dirty firm imposes social cost c_H .
- A green firm imposes lower social cost $c_L < c_H$.
- Reform turns the firm from dirty to green forever at one-time cost $\kappa > 0$.

- The entrepreneur has prosocial preference parameter $\gamma \geq 0$.
- An SRI has prosocial preference parameter $\lambda > 0$.

Interpretation: Reform creates social value but requires whoever controls the firm to pay a private cost. Ownership and bargaining determine whether reform happens immediately or only after acquisition.

2.3 Broad prosocial preferences

The entrepreneur's per-period utility from a firm with social cost $c \in \{c_L, c_H\}$ is

$$u_E = \pi - \gamma c. \quad (1)$$

If she no longer owns the firm, she still suffers disutility $-\gamma c$. Similarly, an SRI suffers disutility $-\lambda c$ whether or not he owns the firm.

Interpretation: Broad preferences mean that agents care about the externality itself, not merely the warm glow of holding a green security. This is what makes acquisition and reform socially meaningful.

2.4 Search frictions

- Time is discrete and infinite.
- Each period, the firm owner may meet one investor.
- The probability of meeting an SRI is N_s/N .
- The probability of meeting an FI is N_f/N .
- If a dirty firm is not reformed by the entrepreneur, it may still be reformed later when an SRI eventually acquires it.

Interpretation: Search frictions create time and make delay meaningful. Without frictions, the dirty firm could immediately move to the party who values reform most.

3 Models and theoretical specifications

3.1 SRI reform condition

If an SRI owns a dirty firm, he reforms it when

$$\frac{\pi - \lambda c_L}{1 - \beta} - \kappa \geq \frac{\pi - \lambda c_H}{1 - \beta}. \quad (2)$$

Equivalently,

$$\lambda \geq \eta_1 \equiv \frac{(1 - \beta)\kappa}{c_H - c_L}. \quad (3)$$

Interpretation: η_1 is the minimum prosocial preference needed to make reform privately optimal for the SRI. Higher reform cost κ raises the threshold; a larger externality reduction $c_H - c_L$ lowers it.

3.2 Benchmark with no SRIs

When there are no SRIs, the entrepreneur reforms if and only if

$$\gamma \geq \eta_1. \quad (4)$$

Interpretation: Without SRIs, there is no dirty-firm resale premium. The entrepreneur simply compares the reform cost with her discounted benefit from lower social cost.

3.3 Why dirty firms become valuable targets

With an FI, the acquisition price depends only on financial profits:

$$P_L = P_H = \pi. \quad (5)$$

With an SRI, a dirty firm has additional acquisition value because the SRI can reform it:

$$P_H = \pi + \lambda(c_H - c_L) - \kappa, \quad P_L = \pi. \quad (6)$$

Interpretation: The dirty firm commands a premium because it gives the SRI an opportunity to generate impact. That premium undermines the current owner's incentive to reform before sale.

3.4 Strategic delay region

The infinite-horizon equilibrium generates thresholds η_1 and η_2 such that strategic delay occurs for

$$\gamma \in [\eta_1, \eta_2). \quad (7)$$

Interpretation: In this region, the entrepreneur is prosocial enough to reform without SRIs, but she does not reform when SRIs are present because the dirty-firm sale option is more valuable.

3.5 Expected delay

If the entrepreneur does not reform, reform occurs when an SRI eventually acquires the firm. The expected delay is

$$T^{\text{delay}} = \sum_{t=1}^{\infty} \left(1 - \frac{N_s}{N}\right)^{t-1} \frac{N_s}{N} t = \frac{N}{N_s}. \quad (8)$$

Interpretation: Conditional on no proactive reform, more SRIs reduce waiting time. But more SRIs also affect the owner's initial incentive to stay dirty, so total welfare effects are not monotone.

3.6 Price-commitment mandate

A price-commitment mandate induces immediate reform by committing SRIs to pay a premium for green firms:

$$\tilde{P}_L = \frac{\pi}{1 - \beta} + \frac{c_H - c_L}{\beta N_s / N} (\eta_2 - \gamma) > \frac{\pi}{1 - \beta}. \quad (9)$$

Interpretation: The mandate flips incentives. Without a mandate, dirty firms receive the premium. With a credible mandate, green firms receive the premium, encouraging reform before sale.

4 Identification and conceptual design

4.1 What the theory explains

- Impact capital can speed up reform by eventually acquiring and reforming dirty firms.
- The same capital can slow reform by making dirty firms more valuable to current owners.
- The key economic object is the timing of reform, not merely whether reform happens at some point.

Interpretation: The mechanism is a timing distortion. A policy that improves eventual reform can still reduce timely reform.

4.2 Why private markets matter

- Private markets involve bilateral search, negotiation, and control transfers.
- Acquisition prices can reflect buyer-specific impact opportunities.
- Private funds often use mandates, making commitment a realistic contractual tool.

Interpretation: The mechanism is strongest when investors can acquire control and when sellers bargain over the price of that control.

4.3 Why free-riding can help

- More SRIs raise the chance of meeting an SRI, which can increase the payoff from staying dirty.
- More SRIs also increase free riding: each SRI can hope another SRI buys and reforms the dirty firm, reducing any one SRI's willingness to pay.
- The two effects push the strategic delay region in opposite directions.

Interpretation: In standard public-good problems, free riding is bad. Here, free riding can lower the dirty-firm premium and reduce strategic delay.

4.4 What the paper cannot prove empirically

- The paper is theoretical and does not estimate delay in real transactions.
- It does not measure actual ESG improvement before versus after impact-fund acquisition.
- It assumes enough commitment power for mandates; real mandates may be incomplete or reputationally enforced.
- It abstracts from heterogeneous technologies, multiple reforms, imperfect green measurement, and regulation.

5 Tables, figures, and original equation panels

The paper is mainly theoretical. The compact visuals below group related original payoff tables, the expected-delay figure, and key proposition/equation panels. The images are directly cropped from the paper.

5.1 Tables 1 and 2: trading game payoffs

is less than the reduction in her disutility from the social cost. Intuitively, the entrepreneur reforms the firm if her pro-social preferences are sufficiently strong.

Green Firm	Entrepreneur	Investor	Dirty Firm	Entrepreneur	Investor
Not Sell	$\pi - \gamma c_L$	0	Not Sell	$\pi - \gamma c_H$	0
Sell	$P_L - \gamma c_L$	$\pi - P_L$	Sell	$P_H - \gamma c_H$	$\pi - P_H$

Table 1: Trading Game with a Financial Investor

Table 2 shows the entrepreneur's and investor's utilities in the trading game with a socially responsible investor with pro-social preference parameter $\lambda > 0$, who reforms the firm post-acquisition in period $t = 1$, that is, $\lambda(c_H - c_L) > \kappa$. If the firm is already green, there are no gains from trade. Intuitively, the most the investor is willing to pay for the firm is its $t = 1$ profit because he cannot further reduce the social cost if he acquires the firm. If the firm is dirty, the highest price the socially responsible investor is willing to pay is given by

$$\pi - P_H - \kappa - \lambda c_L = -\lambda c_H \Leftrightarrow P_H = \pi + \lambda(c_H - c_L) - \kappa.$$

In particular, we have $P_H > \pi$. Intuitively, the investor values not only the firm's profits, but also the reduction in the social cost post-acquisition. The entrepreneur sells the firm because $P_H - \gamma c_L > \pi - \gamma c_H$. The entrepreneur benefits from the price, which exceeds the firm's profit, and from the anticipated reduction in the social cost by the socially responsible investor.

Figure 1: Original payoff-table logic: trading with financial investors versus socially responsible investors.

Read:

- The FI table shows that an FI creates no incremental social surplus: green and dirty firms are valued symmetrically at financial profit.
- The SRI table shows that a dirty firm gives the SRI an opportunity to pay κ and reduce social cost from c_H to c_L .
- The dirty-firm sale row under SRI ownership is the origin of the dirty-firm premium.

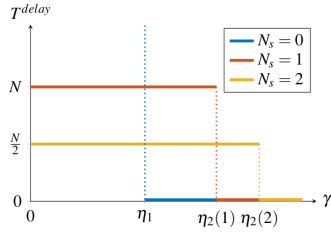
Economic message: With an FI, there is no special payoff from being dirty. With an SRI, being dirty creates an impact opportunity. That opportunity becomes a bargaining surplus and can delay reform.

Interpretation: The key line is “Sell–Dirty firm” under SRI ownership. The entrepreneur receives a high price, while the SRI pays for reform. This allows the entrepreneur to benefit from social-cost reduction without paying the reform cost herself.

5.2 Figure 1 and Proposition 2: expected delay and strategic delay

However, the presence of a socially responsible investor generates strategic delay for an entrepreneur with $\gamma \in [\eta_1, \eta_2(1))$, where $\eta_2(1)$ denotes the threshold in equation (4) when $N_s = \eta_2(1)$, the expected delay is equal to N . Finally, if $N > \frac{\beta}{1-\beta}$, a further increase in N_s increases the strategic delay region but also shortens the expected delay in reform for an entrepreneur who chooses not to reform the firm herself.

Figure 1: Expected Delay T^{delay}



The figure plots the expected delay T^{delay} as a function of the entrepreneur's pro-social preference parameter γ . The horizontal distance between the vertical lines η_1 and $\eta_2(1)$ is the strategic delay region. The vertical distance between the horizontal lines N and $N/2$ is the expected delay in reform for an entrepreneur who chooses not to reform the firm herself.

(a) Original Figure 1: expected delay as a function of γ .

responsive—and net discounted utility is given by $V^1(c_L) - \kappa = \frac{\pi - \gamma c_L}{1-\beta} - \kappa$. Section 4.3.2 shows that if the entrepreneur does not reform the firm, then there are gains from trade in the financial market when the entrepreneur meets a socially responsible investor. In this case, the entrepreneur's expected discounted utility is given by (12). Thus, the entrepreneur reduces the social cost of production in period $t = 0$ if and only if $V^1(c_L) - \kappa \geq V^1(c_H)$, that is, if and only if

$$\frac{\pi - \gamma c_L}{1-\beta} - \kappa \geq \frac{\pi - \gamma c_H}{1-\beta} + \frac{\beta}{1-\beta} \frac{N_s}{N} (P_s(c_H) - \Delta V_s(c_H)). \quad (13)$$

In particular, the entrepreneur's decision to reform the firm depends on the trading surplus $P_s(c_H) - \Delta V_s(c_H) = \Delta Z_s(c_H) - \Delta V_s(c_H)$, which arises endogenously in the financial market.

Proposition 2. *The entrepreneur reforms the firm in period $t = 0$ if and only if*

$$\gamma \geq \eta_1 + \lambda \frac{\beta \frac{N_s}{N}}{1-\beta \left(1 - \frac{N_s-1}{N}\right)} =: \eta_2, \quad (14)$$

where $\eta_1 = (1-\beta) \frac{\kappa}{c_H - c_L}$ is defined in Assumption 1.

The main takeaway from Proposition 2 is that the presence of socially responsible investors in the financial market can lead to a delay in the reduction of the social cost of production. Specifically, the entrepreneur reforms the firm if and only if condition (14) is satisfied. Recall from Section 4.2 that if the financial market is populated only by financial investors, the entrepreneur reduces the social cost of production if and only if $\gamma \geq \eta_1$. This implies that if $\eta_1 \leq \gamma < \eta_2$, the entrepreneur strategically delays the firm's reform in the sense that she does not reform the firm in the presence of socially responsible investors even though she would reform the firm in the absence of socially responsible investors. We therefore refer to $[\eta_1, \eta_2)$ as the strategic delay region.

(b) Original Proposition 2 panel: reform threshold and delay region.

Read:

- With no SRIs, reform is immediate for $\gamma \geq \eta_1$ and never happens for $\gamma < \eta_1$.
- With SRIs, the firm can eventually be reformed after acquisition even if the entrepreneur is weakly prosocial.
- For intermediate γ , SRIs generate strategic delay: the entrepreneur would reform without SRIs but waits in the SRI market.
- The vertical distance in the figure is driven by search probabilities; the horizontal thresholds identify incentive regions.

Economic message: More socially responsible capital can reduce delay after a firm stays dirty, but it can also make the owner choose to stay dirty. The net effect depends on both search speed and the dirty-firm premium.

Interpretation: The figure should not be read as “more SRIs always help.” More SRIs can shorten conditional waiting time, yet the strategic-delay region may expand if sellers expect a higher probability of meeting an impact buyer.

5.3 Pricing and comparative statics

Using $Z_s^1(c_H) = \frac{\pi - \lambda c_L}{1 - \beta} - \kappa$, we get

$$P_s(c_H) = \Delta Z_s(c_H) = Z_s^1(c_H) - Z_s^0(c_H) = \frac{\pi}{1 - \beta} + \frac{\lambda(c_H - c_L)}{1 - \beta \left(1 - \frac{N_s - 1}{N}\right)} - \kappa. \quad (9)$$

We can show that the trading surplus when the entrepreneur meets a socially responsible investor is positive if the pro-social preference parameter of socially responsible investors, λ , is sufficiently high. Formally, we establish the following proposition.

Proposition 1. *There exist values $V^1(c_H)$ and $Z_s^0(c_H)$ and an acquisition price $P_s(c_H)$ satisfying the equilibrium conditions (2), (5), and (6) if and only if*

$$\lambda > (\eta_1 - \gamma) \frac{1 - \beta \left(1 - \frac{N_s - 1}{N}\right)}{1 - \beta}, \quad (10)$$

where $\eta_1 = (1 - \beta) \frac{\kappa}{c_H - c_L}$ is defined in Assumption 1. If condition (10) is satisfied, then the trading surplus when the entrepreneur with a dirty firm meets a socially responsible investor, $\Delta Z_s(c_H) - \Delta V_s(c_H)$, is positive, the entrepreneur sells the firm when she meets a socially responsible investor, and the acquisition price $P_s(c_H)$ is given by (9).

Proposition 1 establishes that equilibrium values $V^1(c_H)$ and $Z_s^0(c_H)$ and an acquisition price $P_s(c_H)$ exist if condition (10) is satisfied.¹² If $\gamma \geq \eta_1$, then condition (10) is always satisfied. If $\gamma < \eta_1$, then condition (10) is satisfied as long as the pro-social preference parameter of socially responsible investors, λ , is sufficiently high. Henceforth, we make the following assumption:

Assumption 2. *Condition (10) is satisfied.*

Proposition 1 further shows that if condition (10) is satisfied, there are gains from trade when

(a) Original Proposition 1 panel: dirty-firm acquisition price and trading gains.

Proposition 3. *Strengthening socially responsible investors' pro-social preferences increases the size of the strategic delay region. An increase in the number of socially responsible investors can increase or decrease the size of the strategic delay region. Specifically, the threshold η_1 does not depend on λ or N_s . The threshold η_2 increases in λ , increases in N_s if $N > \frac{\beta}{1 - \beta}$, and decreases in N_s if $N < \frac{\beta}{1 - \beta}$.*

(b) Original comparative statics panel: how λ and N_s affect delay incentives.

Read:

- Proposition 1 characterizes the dirty-firm price and establishes when trading gains exist with SRIs.
- Stronger SRI prosocial preferences λ raise η_2 , expanding the strategic-delay region.
- The effect of the number of SRIs is ambiguous because search probability and SRI free riding move in opposite directions.

Economic message: The more an SRI values reform, the more a dirty firm can be worth to the seller. This is the paper's strongest paradox: stronger prosocial motives can worsen delay incentives.

Interpretation: The comparative statics distinguish the intensity of prosocial preferences from the quantity of prosocial capital. Increasing λ raises dirty-firm willingness to pay. Increasing N_s raises match probability but may reduce individual SRI willingness to pay through free riding.

5.4 Mandates: exclusionary mandates versus price commitment

Proposition 4. *If $\gamma \in [0, \eta_1)$, there does not exist an exclusionary investment mandate that induces the entrepreneur to reform the firm in period $t = 0$, where η_1 is defined in Assumption 1. If $\gamma \in [\eta_1, \eta_2]$, there exists an individually-rational exclusionary investment mandate through which socially responsible investors commit to not acquiring a dirty firm that induces the entrepreneur to reform the firm in period $t = 0$, where η_2 is defined in Proposition 2.*

Proposition 4 establishes that if $\gamma \in [\eta_1, \eta_2]$, socially responsible investors adopt an investment mandate through which they commit to not acquiring a dirty firm. This result is intuitive. If

(a) Original mandate panel: exclusionary mandates can remove strategic delay only for some owners.

mandate through which they commit to offer a price for a green firm that is greater than the discounted profits, that is, $\bar{P}_L > \frac{\pi}{1-\beta}$. The investment mandate does not include a price offer for a dirty firm.¹⁴ Recall from Section 4.3.1 that in the absence of an investment mandate, the price a socially responsible investor would be willing to pay for a green firm is lower than or equal to $\frac{\pi}{1-\beta}$, which is not sufficient to induce the entrepreneur with $\gamma \in [0, \eta_2)$ to reform the firm. In particular, if $\gamma \in [\eta_1, \eta_2)$, the entrepreneur strategically delays the firm's reform. The problem arises because there are trading gains only if the entrepreneur owns a dirty firm but not if she owns a green firm. By offering to pay a premium for a green firm, socially responsible investors can correct the entrepreneur's incentives and induce the entrepreneur to reform the firm immediately. Importantly, in contrast to an exclusionary investment mandate, the investment mandate with price commitment can also induce an entrepreneur with weak pro-social preferences $\gamma \in [0, \eta_1)$ to reform the firm.

Although there exists a price for a green firm that induces the entrepreneur to reform the firm, it is not immediately clear that such an investment mandate would be adopted in equilibrium because paying a premium for a green firm is costly for socially responsible investors. The price $\bar{P}_L$ is chosen such that the entrepreneur is indifferent between reforming and not reforming the firm. In other words, $\bar{P}_L$ is the minimum price that socially responsible investors must commit to for a green firm to induce the entrepreneur to reform the firm. Importantly, each individual socially responsible investor internalizes that if he deviates to not adopting the investment mandate, the entrepreneur switches to not reforming the firm and selling the dirty firm in the financial market. Since socially responsible investors incur an additional disutility from this delay in each period in which the firm is dirty, they find it individually rational to commit to the investment mandate even though offering a premium is costly. Intuitively, socially responsible investors prefer to commit to

(b) Original price-commitment panel: a green premium can implement immediate reform.

Read:

- An exclusionary mandate not to buy dirty firms can remove the resale premium for moderately prosocial entrepreneurs.
- But exclusion cannot induce reform by weakly prosocial entrepreneurs; if nobody buys dirty firms and the owner does not care enough, the firm stays dirty.
- A price-commitment mandate pays a premium for green firms, making pre-sale reform privately valuable for the entrepreneur.
- The commitment is individually rational for SRIs because it avoids the cost of waiting for reform later.

Economic message: Exclusion removes the reward for dirtiness; price commitment adds a reward for greenness. The latter is stronger because it can induce reform even when the owner would otherwise stay dirty.

Interpretation: The price-commitment mandate makes an apparently passive strategy economically active. The SRI does not need to buy a dirty firm to generate impact; credible future demand for green firms changes today's reform incentives.

5.5 Heterogeneous entrepreneurs

in period $t = 0$ irrespective of her type.

Proposition 5 characterizes when socially responsible investors adopt an investment mandate that induces both entrepreneur types to reform the firm. In particular, they will not always find it optimal to do so. Whether socially responsible investors are willing to adopt the proposed investment mandate depends on the pro-social preference parameter of the high- γ entrepreneur, γ_h , and the probability of drawing a low- γ entrepreneur, w_0 . Specifically, the proposed investment mandate is always individually rational if the high- γ entrepreneur is not too pro-social (i.e., if $\gamma_h < \bar{\gamma}$). If instead the high- γ entrepreneur is sufficiently pro-social (i.e., if $\gamma_h \geq \bar{\gamma}$), the proposed investment mandate is individually rational if and only if the probability of drawing the high- γ entrepreneur is sufficiently low (i.e., if $w_0 \geq \bar{w}_0$).

To understand the intuition for this result, consider a socially responsible investor who unilaterally deviates to not adopting the proposed investment mandate. Recall that the price $\bar{P}_L(0)$ is chosen such that the low- γ entrepreneur is indifferent between reforming and not reforming the firm. As a result, the deviation induces the low- γ entrepreneur to switch to not reforming the firm. The key question is what happens to the high- γ entrepreneur. If the high- γ entrepreneur is not too pro-social (i.e., if $\gamma_h < \bar{\gamma}$), then she also switches to not reforming the firm. Intuitively, if both entrepreneur types switch to not reforming the firm, then this generates a high cost of delay for socially responsible investors and the price commitment investment mandate is always individually rational in this case.

In contrast, if the high- γ entrepreneur is sufficiently pro-social (i.e., if $\gamma_h \geq \bar{\gamma}$), then the high- γ entrepreneur still reforms the firm even if one socially responsible investor deviates from the proposed investment mandate, which introduces a trade-off. On the one hand, if the entrepreneur turns out to be a low- γ entrepreneur, then the deviation induces the entrepreneur to not reform the firm which makes the deviation costly for the socially responsible investor due to the delay

Figure 5: Original panel: investment mandates with multiple entrepreneur types.

Read:

- When entrepreneurs differ in prosocial preferences, a single mandate may not induce all types to reform.
- SRIs may optimally choose a mandate that induces reform by high- γ owners but not by low- γ owners.
- This introduces a screening trade-off: stronger incentives increase reform but may require costly premiums for all green firms.

Economic message: Mandates are powerful but not free. With heterogeneous sellers, the optimal contract may target the marginal group for whom incentives are cheapest to change.

Interpretation: This section is important because it makes the mandate result realistic. In practice, sellers differ in abatement costs, preferences, and ability to reform. A uniform green premium may produce partial, not universal, reform.

6 Short summary

- The paper studies the pace of firm reform in private markets with socially responsible investors.
- SRIs can eventually reform dirty firms after acquisition, but their willingness to pay for dirty targets can make dirty ownership more valuable.
- This creates strategic delay: sellers may postpone reform because they expect to sell dirty firms to SRIs later.
- FIs can exacerbate delay by acting as intermediaries or creating resale channels to SRIs.

- More SRIs shorten expected waiting time conditional on no reform, but their effect on initial reform incentives is ambiguous.
- Exclusionary mandates can remove some delay, but price-commitment mandates are stronger because they reward already-green firms.
- Impact measurement should consider pre-investment changes induced by investor commitments, not only post-investment improvements.

7 Conclusion

7.1 Core conclusion

Gupta, Kopytov, and Starmans show that socially responsible investing can have an unintended timing problem. SRIs can ultimately reduce externalities, but by creating a high acquisition price for dirty firms, they can reduce current owners' incentives to reform early. The market for impact can therefore distort the timing of impact.

7.2 Economic intuition

A dirty firm gives an SRI something to improve. Because the SRI values that improvement, the dirty firm becomes a valuable acquisition target. The entrepreneur can wait, keep the firm dirty, and sell it later. The more strongly the SRI cares about impact, the larger the dirty-firm surplus can be.

7.3 Implications for impact investing

The paper changes how one should evaluate active ownership. Buying dirty firms and improving them after acquisition may look direct, but it may also reward owners for not improving earlier. A green-premium strategy can look less active, yet it may generate faster real change by making reform occur before acquisition.

7.4 Implications for private equity and private credit

Private markets are especially exposed to this mechanism because transactions are negotiated, search is imperfect, and control rights can be transferred. If sellers know that impact buyers pay for dirty assets with reform potential, they may strategically wait rather than invest in abatement.

7.5 Implications for financial intermediaries

FIs matter even when they are not prosocial. They can serve as middlemen who buy dirty assets and resell them to SRIs. Sustainable-finance policy should therefore consider the whole acquisition chain, not only the final impact buyer.

7.6 Implications for mandates and contracting

A mandate is not merely a screen or marketing device; it is a commitment technology. By committing to reward already-green firms, SRIs can change seller incentives before bargaining occurs. Mandate credibility, verification of firm status, and reputational enforcement are central to implementation.

7.7 Implications for measuring impact

If an SRI mandate induces a firm to reform before sale, measured post-acquisition improvement may be zero even though the investment strategy caused the improvement. Impact measurement therefore needs counterfactual timing, not only before-after changes after investment.

7.8 Policy implications

Policies that reward post-acquisition cleanup may increase dirty-firm premia and worsen delay. Policies that reward verified early reform or credible green status can shift incentives toward timely reform. Carbon disclosure, certification, procurement rules, and mandate enforcement can complement SRI capital.

7.9 Research agenda

Future empirical work can test whether dirty assets command premia when impact buyers enter, whether green-premium mandates induce pre-sale reform, whether strict-mandate funds generate more pre-acquisition improvement, and whether intermediary chains delay or accelerate decarbonization.

7.10 Takeaway

Impact is not only about who owns the firm; it is about when incentives change. Active impact investing can solve an externality problem after acquisition, but it can create a timing distortion before acquisition. The most effective strategy may be one that turns future investor demand into a present incentive for firms to clean up sooner.